

Name _____ Period _____ Test Day _____

Unit 7: Ionic & Metallic Bonding

Essential Knowledge and Guided Notes

Textbook: CK Chemistry Chapters 7, 8 & 9

<https://flexbooks.ck12.org/cbook/ck-12-chemistry-flexbook-2.0/>

Practice Ionic Bonding Quizzes from Science Geek: <https://tinyurl.com/GeekIB1>
<https://tinyurl.com/GeekIB2>

Helpful Videos: Posted on CANVAS

Essential Knowledge:

Relationship of Valence Electrons and Charge Remember the Octet Rule (Everyone wants 8)

1	2	The charge on the representative elements can be predicted by the number of valence electrons →										13	14	15	16	17	18
1	2											←	number of valence electrons	→	3	4	5
+1	+2	Transition Elements cannot be predicted. D ORBITALS MAKE IT MORE COMPLICATED										+3		- 3	- 2	- 1	0
+1	+2											3	4	5	6	7	8
+1	+2											+3		- 3	- 2	- 1	0

- The charge on the representative elements can be predicted by knowing the number of valence electrons and remembering the octet rule (8 electrons: ns^2np^6)
- Cations** are formed by **oxidation** which is the loss of electrons. Metals become cations when they lose electrons
- Anions** are formed by **reduction** which is the gain of electrons. Non-metals become anions when they gain electrons
- A way to remember this is **OIL RIG. Oxidation is loss and Reduction is gain.**

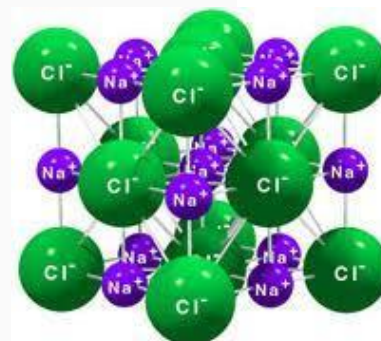
- Lewis Dot Diagrams** show only the valence electrons

1	2	13	14	15	16	17	18
H•							He••
Li•	•Be•	•B•	•C•	•N•	•O•	•F•	•Ne••
Na•	•Mg•	•Al•	•Si•	•P•	•S•	•Cl•	•Ar••
K•	•Ca•				•Se•	•Br•	•Kr••
Rb•	•Sr•				•Te•	•I•	•Xe••
Cs•	•Ba•						

Electron dot configurations for sodium and chlorine atoms



Electron dot configurations for sodium and chloride ions



Remember, NaCl only exists as a cube, with six chloride ions surrounding every sodium, and six sodium ions surrounding every chlorine.

6. Bonds form between atoms to achieve **stability**.
7. **Ionic bonds** involve the transfer of electrons between ions. Elements with low ionization energy (metals) form positive ions (cations) easily. Elements with high ionization energy (non-metals) form negative ions (anions) easily. The oppositely charged ions are then attracted to each other and held together by **electrostatic interactions which are very strong**. These extremely strong bonds result in solids with very high melting points. They can conduct electricity when melted or dissociated (dissolved in water).
8. **Chemical formulas** are used to represent ionic compounds. Subscripts represent the relative number of each type of atom in an ionic formula unit. **In an ionic compound, a formula unit is the smallest whole number ratio of cations to anions in which the charges equal zero.**

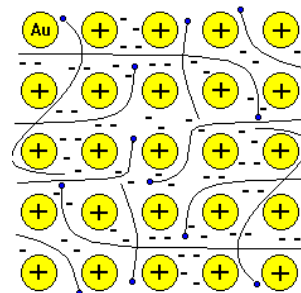
$$\text{AlCl}_3 \text{ (Al}^{3+} \text{ Cl}^{1-} \text{ Cl}^{1-} \text{ Cl}^{1-} = 0) \quad \text{not } \text{Al}_2\text{Cl}_6$$
9. Rules must be followed to name ionic compounds:
 - a. An ionic compound is electrically neutral. In other words, the positive electrical charges of the cations exactly balance the negative electrical charges of the anion.
 - b. Write the element name of the cation first. This is almost always a metal. One significant exception is the ammonium cation (NH₄⁺), which is a polyatomic cation consisting of nonmetals. It forms ionic compounds, such as ammonium chloride (NH₄Cl).

- c. If the cation element has more than one possible oxidation state (charge) follow the element name using parentheses containing the appropriate Roman numeral. The elements this applies to are some transition metals. For example, the iron cation is either iron(II) or iron(III). Metals in group 1 and group 2 always go by their element name, with no Roman numerals.

Common Polyvalent Ions (for use w/Roman Numerals)	
Cu	+1, +2
Hg	+1, +2
Fe	+2, +3
Mn	+2, +3
Co	+2, +3
Cr	+2, +3
Sn	+2, +4
Pb	+2, +4

- d. The anion name follows the cation. A space separates the two parts of the ionic compound name. If the compound is a binary compound or the anion is a single type of atom, the anion name is a combination of the element name with the -ide ending. For example, AgCl has the name silver chloride.
- e. If the anion consists of multiple types of atoms, it is called a polyatomic ion and you must use its polyatomic name. Here is a link to a list of common polyatomic ions: <https://sciencenotes.org/polyatomic-ions-list> **You will be expected to memorize the following 6 polyatomic ions: NH_4^{1+} (ammonium), OH^{1-} (hydroxide), NO_3^{1-} (nitrate), SO_4^{2-} (sulfate), CO_3^{2-} (carbonate) and PO_4^{3-} (phosphate). SEE PAGE 10 FOR AN EASY WAY TO MEMORIZE THEM.**

10. A metallic bond forms when a metal cation attracts freely moving delocalized electrons. In the **electron sea model**, electrons move through the metallic crystal and are not held by any particular atom. It explains the properties of a metal such as heat and electrical conductivity, ductility, and malleability. Alloys are mixtures of metals. Common Alloys are Bronze (90% copper, 10% iron), Brass (70% copper, 30% zinc), and Stainless Steel (74% Iron, 18% Chromium, 8% Carbon).



11. Properties of Ionic Compounds vs. Covalent Compounds Summary

Ionic Compounds:

- Consist of a metal cation (or ammonium) **transferring** electrons to non-metal anions
- Are a solid array of ions.
- Have a high melting point.
- Conduct electricity when dissociated in water (electrolytes)

Covalent Compounds:

- Consist of TWO or more non-metals **SHARING** electrons
- Are discrete molecules
- Can be solid (ex: sugar), liquid (ex: water) or gas (ex: CO_2)
- Do not conduct electricity when dissolved in water. (non-electrolytes)

U7 Lesson 1: Ionic Bonding and Nomenclature

1. Ions are formed to help elements achieve _____. There are three types of bonding that occur: _____, _____ and _____. We will begin with ionic bonding.
2. Look at each correctly written formula for these ionic compounds.

What do you notice about the way each compound formula is written? (Hint: Look at the types of elements and the location on the PT)

Formulas

Observations:

LiCl
NaF
BaO
MgS
AlN
Na₂S
AlF₃
CaI₂

3. Look at the following correctly written names for the formulas above you observed with your group. What do you notice about the way each compound name is written?

Names

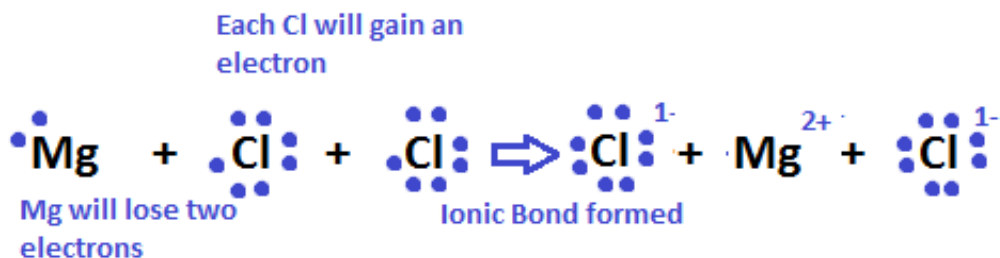
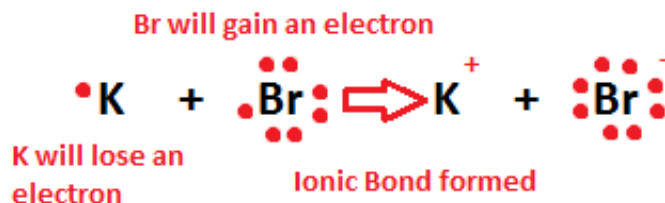
Observations:

lithium chloride
sodium fluoride
barium oxide
magnesium sulfide
aluminum nitride
sodium sulfide
aluminum fluoride
calcium iodide

4. Lewis Dot Diagrams show _____ electrons. They help us visualize bonds. Practice writing the following Lewis Dot Diagrams with your group. Be ready to check your answers:

H Na O S N Ne F He

5. Ionic compounds are formed by the _____ between positive cations and negative anions. Charges on the ions in an ionic formula must add up to _____ for the compound to be stable. This is because just as many electrons must be _____ by metals through oxidation as _____ by non-metals by reduction for the compound to be _____.



6. Observe how the formation of ionic bonds is visualized with Lewis Dot diagrams above. Draw the formation of an ionic compound between Al and N in the same manner in the space below:

7. Steps for writing an ionic formula
- Determine the charge on the _____. Write the _____ first.
 - Determine the charge on the _____. Write the _____ second.
 - Determine the number of cations and anions needed to make the charge equal to _____.
 - Write the formula with that ratio using _____.
 - Always use the _____ ratio for the subscripts
 - Example: What is the formula between K and P? (potassium phosphide)

8. Use the **criss-cross method ("Swap and Drop")** to find the following formulas:

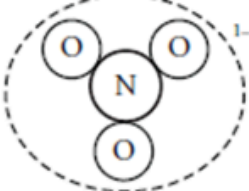
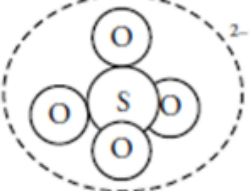
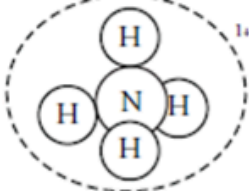
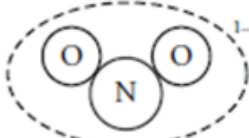
- What is the formula between calcium and nitrogen? _____
 - What formula between K and S? _____
 - What is the formula between Mg and S? _____
- Be careful, formulas should be the lowest ratio.*

9. **Naming Rules: Ionic Compounds only have two words in their names**

Nomenclature: Binary ionic compounds (2 elements)

- The "First" name is the Cation (usually a metal)
- The "Last" name is the Anion (always a non-metal or a group containing non-metals)
- When naming ionic compounds, the metal is written first followed by the non-metal whose ending is changed to "ide"
- Example: CaF_2 calcium fluoride
- Al_2O_3 _____
- Na_2S _____

Model 1 – Types of Ions

Monatomic Ions	Nitride 	Sulfide 	Chloride 
Polyatomic Ions	Nitrate 	Sulfate 	Ammonium 
	Nitrite 	Sulfite 	Hydroxide 

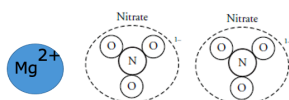
10. **Nomenclature:** Ternary ionic compounds (3+ elements)

- Ternary compounds always contain a group of nonmetals that are covalently bonded but carry a charge. These groups are called _____ ions.
- Look at the model above. What are your observations about the polyatomic ions vs the monatomic ions?

- Look at the formulas below: What are your observations about the formulas of the polyatomic ions?

Monatomic Ions	N^{3-} nitride	O^{2-} oxide	F^{1-} fluoride
Polyatomic Ions	NO_3^{1-} nitrate	SO_4^{2-} sulfate	NH_4^{1+} ammonium
	NO_2^{1-} nitrite	SO_3^{2-} sulfite	OH^{1-} hydroxide

Picture of Magnesium Nitrate



Formula = $\text{Mg}(\text{NO}_3)_2$

- What do you notice about the name and formula?

- e. Compounds containing polyatomic ions still only have two words in their names.

Common Polyatomic Ions			
$\text{C}_2\text{H}_3\text{O}_2^-$	acetate	OH^-	hydroxide
NH_4^+	ammonium	ClO^-	hypochlorite
CO_3^{2-}	carbonate	NO_3^-	nitrate
ClO_3^-	chlorate	NO_2^-	nitrite
ClO_2^-	chlorite	$\text{C}_2\text{O}_4^{2-}$	oxalate
CrO_4^{2-}	chromate	ClO_4^-	perchlorate
CN^-	cyanide	MnO_4^-	permanganate
$\text{Cr}_2\text{O}_7^{2-}$	dichromate	PO_4^{3-}	phosphate
HCO_3^-	bicarbonate	SO_4^{2-}	sulfate
HSO_4^-	bisulfate	SO_3^{2-}	sulfite
HSO_3^-	bisulfite		

Naming Examples:

- NH_4Cl is ammonium chloride
- Na_3PO_4 is sodium phosphate
- Name $\text{Al}(\text{NO}_3)_3$ _____
- Name KNO_3 _____
- Name $\text{Mg}_3(\text{PO}_4)_2$ _____
- Write the formula: sodium sulfate _____
- Write the formula: ammonium sulfite _____

1	=	I
2	=	II
3	=	III
4	=	IV
5	=	V
6	=	VI
7	=	VII
8	=	VIII
9	=	IX
10	=	X

11. **Nomenclature:** Multivalent Cations

Some transition metals such as iron, copper, nickel, and manganese have more than one common charge or oxidation state. Some larger "p block" metals do as well. To indicate the charge of the ion Roman numerals are used.

$\text{Fe(II)} = \text{Fe}^{2+}$ and $\text{Fe(III)} = \text{Fe}^{3+}$

- Name: CuO _____ () _____
- Name: Cr_2S_3 _____ () _____
- Formula: iron (III) chloride _____
- Formula: lead (IV) sulfide _____

Common Polyvalent Ions (for use w/Roman Numerals)

Cu	+1, +2
Hg	+1, +2
Fe	+2, +3
Mn	+2, +3
Co	+2, +3
Cr	+2, +3
Sn	+2, +4
Pb	+2, +4

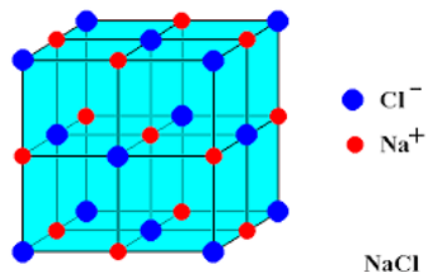
12. **Important Note:**

- Since group 1 and group 2 metals ONLY have 1 charge (1+ and 2+) they NEVER include roman numerals.
- Two other very important metals that only have 1 charge and therefore do not need a roman numeral are **Zinc: Zn^{2+} and Silver: Ag^{1+}**
- Assume all other transition metals (and metals in groups 14 & 15) need roman numerals**

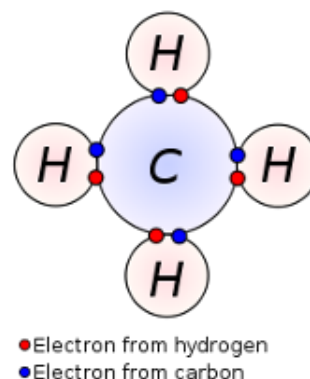
U7 Lesson #2: Ionic vs. Covalent Bonding

1. Elements bond in compounds to achieve _____.

a. An ionic bond is formed when a metal _____ electrons to form a _____ and a non-metal _____ electrons to form an _____. The ions are held together in crystals by _____ attraction which is very strong. All ionic substances are _____. **Ionic Compounds usually contain metal cations and non-metal anions(unless the cation is ammonium, NH_4^{1+}).**



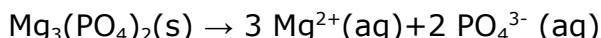
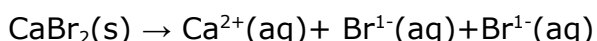
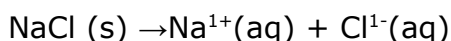
b. A covalent bond is formed when a non-metal _____ electrons with another non-metal to achieve an _____ and form _____. These molecules have differing attractions for each other. Covalent molecular compounds can be _____, _____ or _____ depending on the strength of the attraction between the molecules. **Covalent compounds contain all non-metal atoms**



2. The behavior of Ionic and Covalent compounds in water is different.

a. When you put ionic compounds in water, they _____ into ions that are charged.

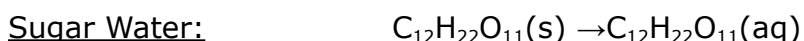
Examples of the dissociation of ionic compounds:



What do the (s) and (aq) stand for?

b. When you put covalent compounds in water, they _____ into uncharged molecules

Examples of the dissolving of covalent compounds:



3. Ionic and Covalent compounds act differently in water when conducting electricity.

To conduct electricity you need moving charged particles.

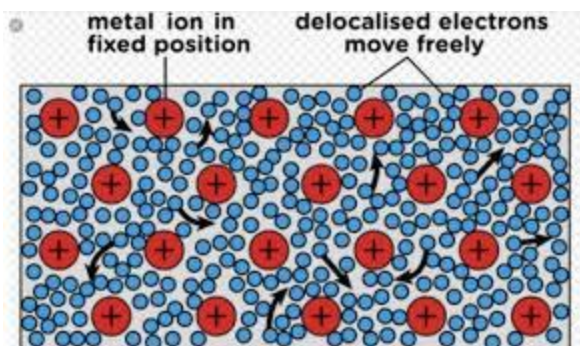
a. Solutions that contain _____ are called _____ because they can conduct an electric current. **Ionic Compounds form electrolytes when they dissociate into ions in water.**

b. Solutions that **do not contain** _____ are called _____ because they cannot conduct an electric current. **Covalent Compounds form non-electrolytes when they dissolve into neutral uncharged molecules in water**

Unit 7 Lesson #3: Metallic Bonding

Video: What is a metal? (Metallic bonds)

1. Metals are _____, _____, _____, _____ and _____
2. The model that is used to describe metallic bonding shown here is called a _____ of mobile valence electrons



3. Metallic bonds are _____ than ionic and covalent bonds.
4. List 4 Properties described in the video that are explained by this model:
 - a.
 - b.
 - c.
 - d.

(Note, pay close attention to how the sea of mobile electrons model explains each of these properties.)

5. What is the only metal that is not a solid at room temperature? What phase is it?

6. What are the 4 metals that melt very close to room temperature?

a.

b.

c.

d.

EASY WAY TO MEMORIZE THE SIX REQUIRED POLYATOMICS

OH NO SO CO PO and NH